

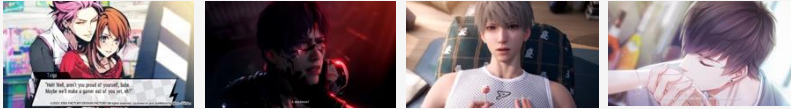
Across Virtual and Real Worlds: Attachment and Boundary Permeability in Relationships with Virtual Characters

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Introduction

Parasocial relationships: One-sided emotional bonds with mediated figures, despite the absence of reciprocal real-world interaction (Horton & Wohl, 1956).

Dating-simulation (otome) games: A type of game that allows players to engage in interactive narratives in which they “date” virtual characters, experiencing plots and dialogues typically associated with real-life romantic partners.



Within player communities, a recurring debate often emerges: *“Would you want the game character to become a real person and date you?”* While some players express a strong desire for this transformation, others firmly reject it, suggesting a divergence in players’ interpretations of their relationships with characters, as well as of the boundary between virtual and real relational domains.

Attachment theory provides a framework for understanding such divergence. Individuals high in attachment anxiety tend to exhibit hyperactivation and proximity-seeking strategies under relational threat, whereas individuals high in attachment avoidance adopt deactivation and distancing strategies instead (Mikulincer & Shaver, 2003).

Prior research was mainly focused on the formation of parasocial attachment and understanding the parasocial involvement and intimacy with mediated figures (Feng, 2025; Cole & Leeds, 1999; Greenwood & Long, 2011). However, little is known about how individuals interpret the nature of these relationships, particularly in romantic video game contexts, and how attachment styles shape the perceived permeability between virtual and real relationships.

Hypotheses

Anxious attachment will predict (+)

-> **Greater permeability** across virtual and real boundaries

Avoidant attachment will predict (-)

-> **Lower permeability** across virtual and real boundaries

Boundary permeability indicators:

- **Realism belief:** Belief of the character as “real”
- **Game Presence:** Cognitive perception of the presence of the character as
- **Desire for realism:** Desire for the character to exist in real world
- **Real-world engagement** with game characters
- **Relationship dependency**

Participants

273 dating-simulation game players

- Female: 89.9%
- Mean age: 25.6

Predictors

Attachment toward game characters

Measured with the *Experiences in Close Relationships* (ECR) scale (Brennan et al., 1998)

(1) **Anxiety** (18 items)

e.g., “I worry a lot about my relationships with [character].”

(2) **Avoidance** (18 items)

e.g., “I prefer not to be too close to [character].”

Outcomes

Boundary Permeability Indicators

(1) **Realism belief** (1 item)

“I believe [character] is real.”

(2) **Game Presence**

Temple Presence Inventory (11 items) (Lombard et al., 2009)

e.g., “How often did you have the sensation that people you saw/heard in the game could also see/hear you?”

(3) **Desire for realism**

5 items; $\alpha = 0.92$

e.g., “I wish [character] could become a real person.”

(4) **Real-world engagement**

8 items; $\alpha = 0.94$

e.g., “How often do you engage in celebration-related activities for [character] (e.g., celebrating his birthday, your in-game anniversary, or other special dates)?”

(5) **Relationship dependency**

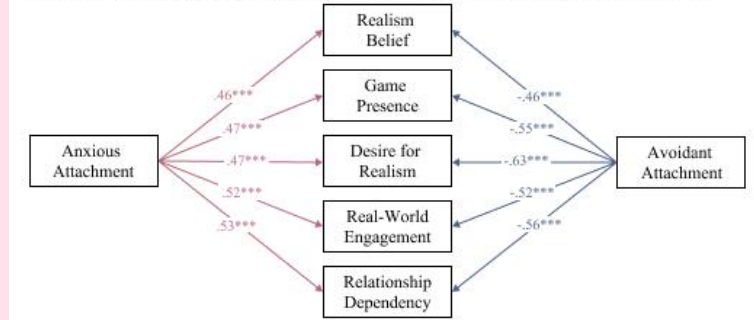
4 items; $\alpha = .83$

e.g., “You feel like you need this relationship with [character] a lot.”

Results

Higher attachment anxiety was positively associated with all indicators of boundary permeability, whereas attachment avoidance was negatively associated with these outcomes (all p-values < .001).

Pearson correlations (r) between attachment dimensions and boundary permeability indicators



Discussion

Extend attachment theory into parasocial relationships in virtual contexts.

- Higher anxiety -> Proximity-seeking -> Extension of virtual relationships into the real world.
- Higher avoidance -> Distancing -> Resistance to permeability of the game into offline life.

Limitations

- Attachment anxiety and avoidance were examined using separate bivariate correlations, which do not isolate their unique effects.
- The extent to which participants conceptualize their relationships as romantic versus entertainment-based was not directly assessed.

Future Directions

- Examine how specific game design features (e.g., interactivity vs. narrative focus; first person vs. third person) moderate attachment-related effects.
- Refine and expand the operationalization of boundary permeability to more comprehensively capture the construct.